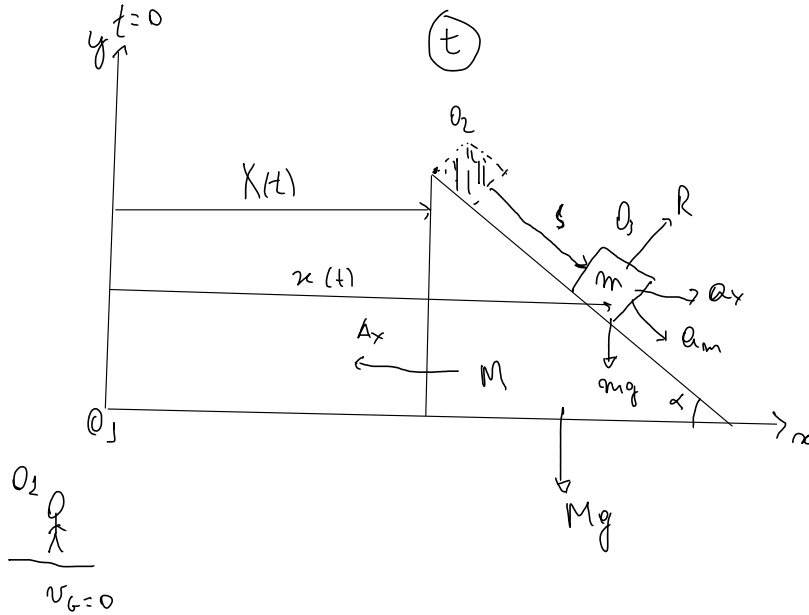




$$a_y = (A_x - a_x) \tan \alpha \quad (1)$$

$$a_x = \frac{Mg \sin \alpha \cos \alpha}{M + m \sin^2 \alpha} \quad (2)$$

$$a_y = -\frac{(M+m)g \sin^2 \alpha}{M + m \sin^2 \alpha} \quad (3)$$

$$A_x = -\frac{m g \sin \alpha \cos \alpha}{M + m \sin^2 \alpha} \quad (4)$$

$$A_y = 0 \quad (5)$$

$$a_m = \frac{(M+m)g \sin \alpha}{M + m \sin^2 \alpha} \quad (6)$$

0,

$$\rightarrow i) M=0 \Rightarrow a_y = -g, a_x = 0, A_x = A_y = 0$$

$$ii) M \gg m \Rightarrow a_y = -g, a_x = g \sin \alpha \cos \alpha, A_y = 0, A_x = 0$$

$$iii) M \ll m \Rightarrow a_y = -g, a_x = 0, A_y = 0, A_x = -g \cot \alpha$$

$$ii) M \gg m \Rightarrow a_m = \frac{g}{\sin \alpha}$$

$$iii) M \ll m \Rightarrow a_m = \frac{g}{\sin \alpha}$$

$$\lim_{\frac{m}{M} \rightarrow 0} a_y = \lim_{\frac{m}{M} \rightarrow 0} -\frac{(1 + \frac{m}{M})g \sin^2 \alpha}{1 + \frac{m}{M} \sin^2 \alpha}$$

$$\lim_{\frac{m}{M} \rightarrow 0} a_y = -g$$

$$\lim_{\frac{m}{M} \rightarrow 0} a_x = \lim_{\frac{m}{M} \rightarrow 0} \frac{g \sin \alpha \cos \alpha}{1 + \frac{m}{M} \sin^2 \alpha}$$

$$\lim_{\frac{m}{M} \rightarrow 0} a_x = g \sin \alpha \cos \alpha$$

$$\lim_{\frac{m}{M} \rightarrow 0} A_x = \lim_{\frac{m}{M} \rightarrow 0} -\frac{m}{M} \frac{g \sin \alpha \cos \alpha}{1 + \frac{m}{M} \sin^2 \alpha}$$

$$\lim_{\frac{m}{M} \rightarrow 0} A_x = 0$$